

STACKR Prototype Improvement Instructions

I really like the current **STACKR prototype**, but I would like to make it more **direct, intuitive, simple, and beginner-friendly**.

Our professor advised us to consider the experience of a **normal, non-technical, or first-time user** who has never used STACKR before. The user should be able to use the system independently without needing someone to explain what to do.

After comparing STACKR with the **ASPECT tool I attached**, I noticed that ASPECT has a much more straightforward flow:

Upload Dataset → Configure/Confirm Inputs → Run System → View Results

Please apply a similar approach to STACKR while preserving its important functionality and thesis requirements.

1. Simplify the Interface

Review the current tabs, sections, metrics, and features. Keep the features necessary for the main workflow and SOP, but move unnecessary or highly technical information into **Advanced/Technical Details** or expandable sections.

The main interface should clearly answer:

- What do I upload?
- What do I need to do?
- What is the system doing?
- What are my results?
- What do the results mean?

A suggested flow is:

Home → Dataset → Configuration → Processing → Results → Loading/Unloading Guide

Please use your judgment to create the simplest structure based on the existing prototype.

2. Make Dataset Import Beginner-Friendly

When the user selects **Add Dataset / Import Your Own Dataset**, show a short explanation of the required dataset format.

Clearly state the required columns, dimensions, data types, and other conditions based on the actual STACKR system and SOP.

Do **not** add a **Destination** column. Our thesis focuses on multi-stop delivery but does not assign specific geographic destinations.

If the dataset contains a **Stop Number, Stop Sequence, or Delivery Sequence**, use that information instead.

If the dataset is invalid, clearly show:

Invalid Dataset

The uploaded dataset does not meet the required format or contains invalid data.

Provide:

Discard Dataset → remove it and allow another dataset to be uploaded

Cancel → cancel the import process

The user should not have to restart STACKR.

3. Simplify the Results

Create a clear **main Results page** that shows the important outputs required by the SOP.

STACKR uses **four algorithm configurations**, so users should be able to:

View Solution → **View Loading/Unloading Arrangement** → **Export Guide**

Keep technical information accessible, but do not show everything at once.

The system should also clearly identify the **Recommended Loading Solution** based on the predefined evaluation criteria in the SOP, such as:

- Space Utilization
- Constraint Satisfaction
- Computational Cost
- Robustness

Briefly explain why it is recommended without overwhelming the user with mathematical or algorithmic details.

Also, completely remove the “**How the solution improved over time**” feature.

4. Add an Exportable Loading and Unloading Guide

After generating a solution, add an **Export Loading Guide** option.

The guide should be practical and printable, allowing a worker to understand how to physically load and unload the packages.

Include information available from the actual STACKR solution, such as:

- Loading Order
- Unloading Order
- Box/Package ID
- Dimensions (L × W × H)

- Weight
- Fragility Level
- Position/Coordinates
- Stop Sequence/Priority
- Stacking or Placement Information
- Weight Capacity Status
- Static Stability/Balance Status
- Base-Support Status
- Space Utilization

Do not include geographic destinations.

5. Include a Simple Visual Arrangement

If the generated solution contains enough positional information, include a simple **2D or 3D container arrangement** in the guide.

The visual should clearly show where each Box ID is placed and should prioritize readability over technical complexity.

For example:

Box 001 → Position 1 → Load First

Box 002 → Position 2 → Place Above Box 001

Box 003 → Position 3 → Keep Accessible for Earlier Unloading

Only show placement instructions that can actually be determined from STACKR's generated solution.

6. Suggested Export Structure

Keep the exported guide simple:

Page 1: Loading & Unloading Guide

Recommended solution, loading/unloading sequence, Box IDs, placement instructions, package properties, and stop sequence.

Page 2: Container Arrangement

Simple visual showing Box IDs and positions.

Page 3: Solution Summary

Space Utilization, Constraint Satisfaction, Weight Capacity, Static Stability/Balance, and other SOP-required metrics.

Optional: Technical Details

Algorithm configuration, computational cost, and other research information.

7. Overall Goal

The final STACKR prototype should feel like a **guided and practical system**, not a complicated technical dashboard.

The intended workflow is:

Upload Dataset → Run STACKR → View Four Algorithm Solutions → Identify Recommended Solution → View Loading/Unloading Arrangement → Export/Print Guide

The system should ultimately communicate:

“Here is the generated solution, here is why it meets the required criteria, and here is a practical guide you can use to follow the loading and unloading arrangement.”

Please preserve the existing **STACKR branding and core functionality**, but prioritize **simplicity, clarity, beginner-friendly instructions, practical usability, and alignment with the thesis SOP**.

The system must remain within the scope of:

Constrained Three-Dimensional Container Loading for Multi-Stop Delivery via Sequential and Repair-Based Intra-Variant Hybrid Grey Wolf Strategies.